

HW-10

Ans-01

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Adjacency List

$$\text{adj}(1) = [2, 3]$$

$$\text{adj}(2) = [4, 5]$$

$$\text{adj}(3) = [6, 7]$$

$$\text{adj}(4) = []$$

$$\text{adj}(5) = []$$

$$\text{adj}(6) = []$$

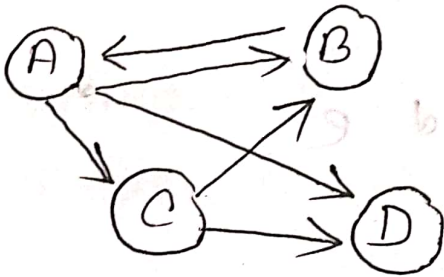
$$\text{adj}(7) = []$$

Adjacency Matrix

	1	2	3	4	5	6	7
1	0	1	1	0	0	0	0
2	0	0	0	1	1	0	0
3	0	0	0	0	0	1	1
4	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0

Ans-05

(a)



(b) Adjacency Matrix:

	A	B	C	D
A	0	1	1	1
B	1	0	0	0
C	0	1	0	1
D	0	0	0	0

Ans-02

Step 01: Output: S

Queue: a, c, d

Step 02: Output: S, a

Queue: c, d

Step 03: Output: S, a, c

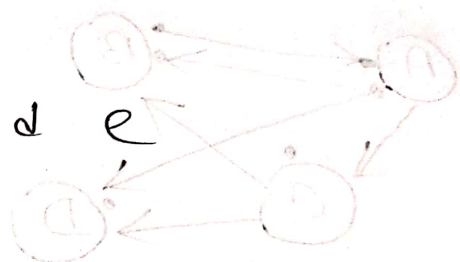
Queue: d, e, b

Step-04: output: ~~sa~~ s a e d

Queue: e b

Step 05: output: s a e d e

Queue: b



Step 06: output: s a e d e b

Queue:

Ans-03

$$\text{Adj}(0) = [1, 2, 3, 7]$$

$$\text{Adj}(1) = [0, 2, 4, 7]$$

$$\text{Adj}(2) = [0, 1, 6]$$

$$\text{Adj}(3) = [0, 5]$$

$$\text{Adj}(4) = [1, 6]$$

$$\text{Adj}(5) = [3]$$

$$\text{Adj}(6) = [2, 4]$$

$$\text{Adj}(7) = [0, 1]$$

BFS:

PO-200

① output: 2

Queue: 0, 1, 6

② output: 2, 0

Queue: 1, 6, 3, 7

③ output: 2, 0, 1

Queue: 6, 3, 7, 4

④ output: 2, 0, 1, 6

Queue: 3, 7, 4

⑤ output: 2, 0, 1, 6, 3

Queue: 7, 4, 5

⑥ output: 2, 0, 1, 6, 3, 7

Queue: 4, 5

⑦ output: 2, 0, 1, 6, 3, 7, 4

Queue: 5

⑧ output: 2, 0, 1, 6, 3, 7, 4, 5

Queue:

Ans-04

Step 01: visited: 0

Stack: 2, 1

Step 02: visited: 0, 2

Stack: 6, 1

Step 03: visited: 0, 2, 6

Stack: 1

Step 04: visited: 0, 2, 6, 1

Stack: 7, 4

Step 05: visited: 0, 2, 6, 1, 7

Stack: 4

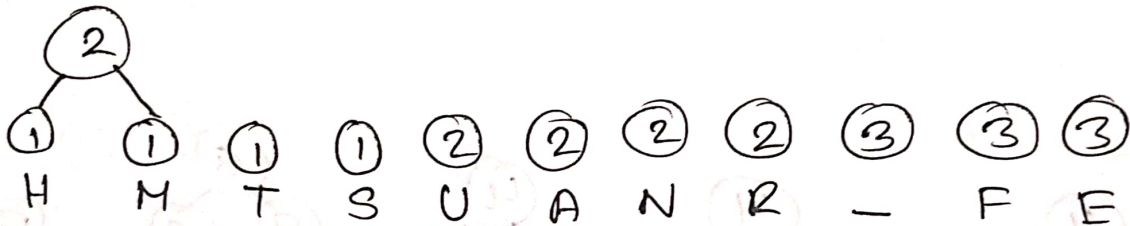
Step 06: visited: 0, 2, 6, 7, 1, 4

Stack:

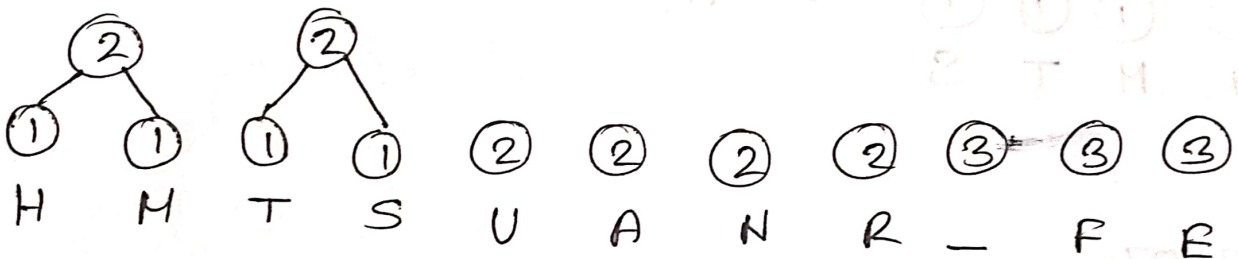
Ans-06

① ① ① ① ② ② ② ② ③ ③ ③
H M T S U A N R - F E

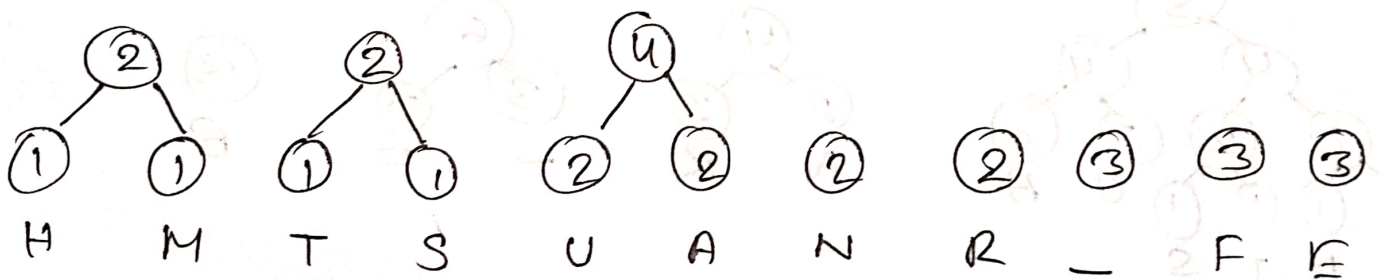
Step-01:



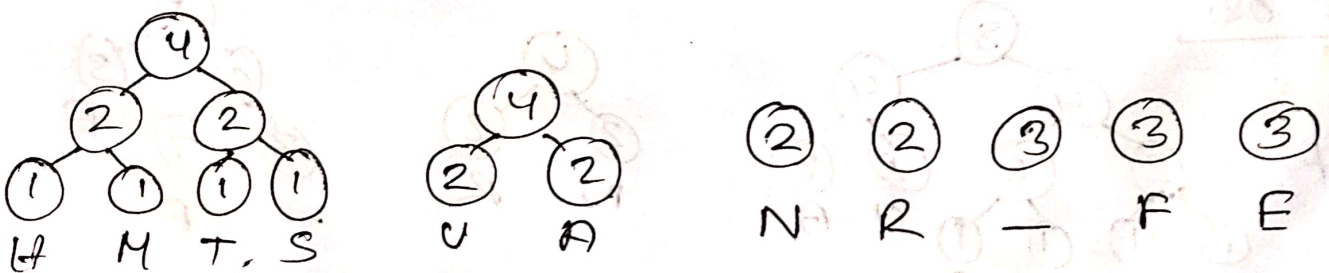
Step-02:



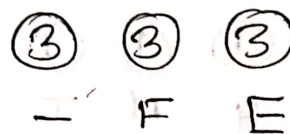
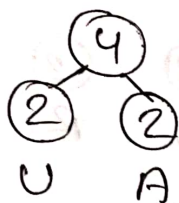
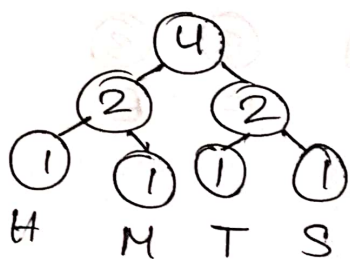
Step-03:



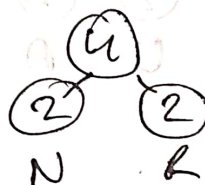
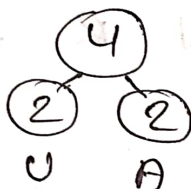
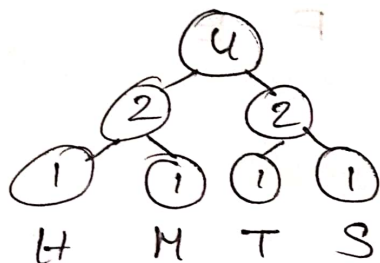
Step-04:



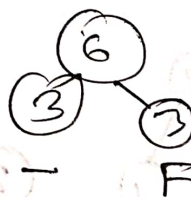
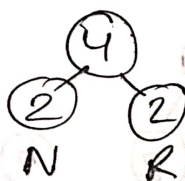
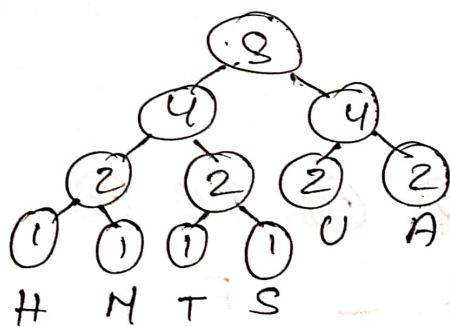
Step 05:



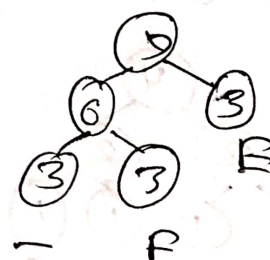
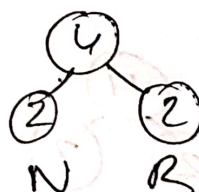
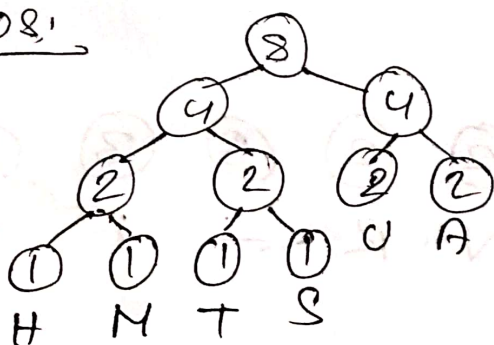
Step 06:



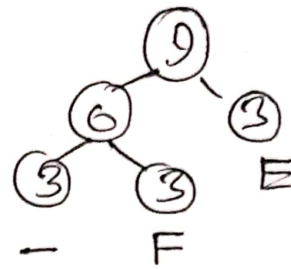
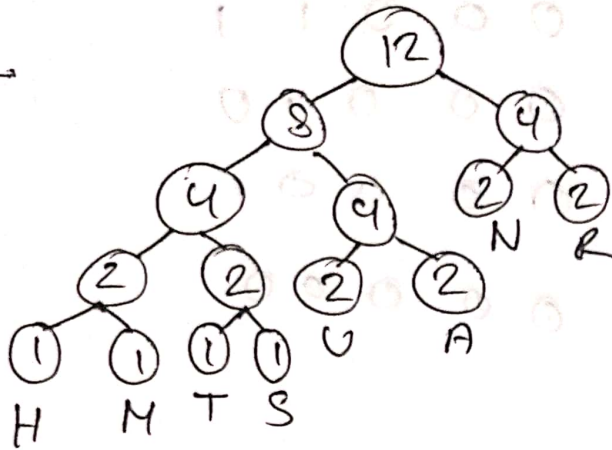
Step 07:



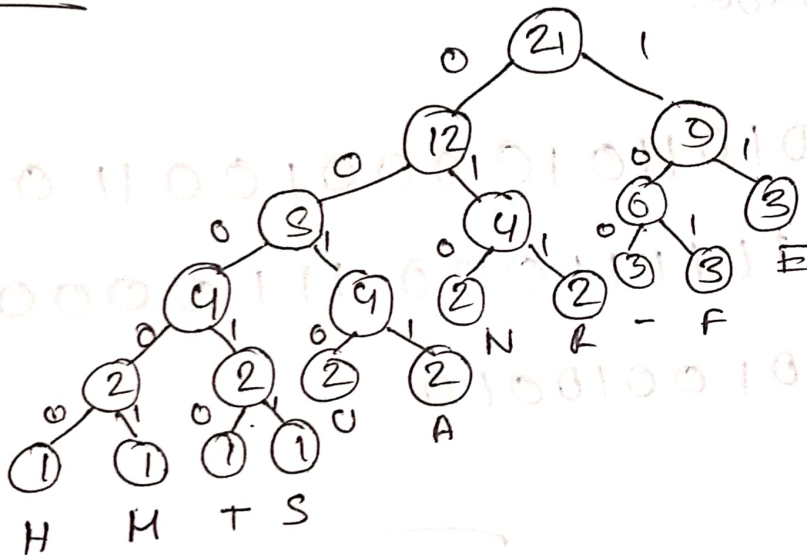
Step 08:



Step-09:



Step-10:



Huffman code:

E	11
F	101
-	100
R	011
N	010
A	0011
U	0010

S

T

M

H

0 0 0 1 1

0 0 0 1 0

0 0 0 0 1

0 0 0 0 0

Encoded Message:

00000000101011010000100110101
00000100111110001110000011
0111001010010010